

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$$([g,h])^\circ = (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$$

the converse of a fold over decimals undoes the algebra $[g,h]$ and then either **wrap**, or the fold's converse on the front and **snoc**

1	$([g,h])^\circ$	$(\emptyset) = [\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}(\emptyset)) \emptyset$
2	$= ([\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}([g,h])) [g,h])^\circ$	$(R + S) [P,Q] = [R P, S Q]$
3	$= ([\text{wrap}, \text{snoc}]^\circ [g, \neg \text{Digit}([g,h]) h])^\circ$	$[U,V]^\circ [R,S] = (U^\circ R) \cup (V^\circ S)$
4	$= ((\text{wrap}^\circ g) \cup (\text{snoc}^\circ \neg \text{Digit}([g,h]) h))^\circ$	$(R \cup S)^\circ = S^\circ \cup R^\circ$
5	$= (\text{snoc}^\circ \neg \text{Digit}([g,h]) h)^\circ \cup (\text{wrap}^\circ g)^\circ$	$R \cup S = S \cup R$
6	$= (\text{wrap}^\circ g)^\circ \cup (\text{snoc}^\circ \neg \text{Digit}([g,h]) h)^\circ$	$(R S)^\circ = S^\circ R^\circ$
7	$= (g^\circ \text{ wrap}) \cup ((\neg \text{Digit}([g,h]) h)^\circ \text{ snoc})$	$(R S)^\circ = S^\circ R^\circ$
8	$= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ \text{ snoc})$	$\neg \text{Digit}(R)^\circ = \neg \text{Digit}(R^\circ)$
9	$= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$	

$\text{val}^\circ = (\text{embed}^\circ \text{ wrap}) \cup (\text{op}^\circ \neg \text{Digit}(\text{val}^\circ) \text{ snoc})$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(6.1b)

1	$\text{op}^\circ(m,p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$ op° splits a number into its quotient and remainder by 10
2	$(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m,p)) \iff 10 \leq m$ op° gives a pair with a nonzero first component exactly at the numbers with two or more digits
3	$(\exists d. \text{embed}^\circ(m,d)) \iff 0 < m \wedge m < 10$ embed° gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

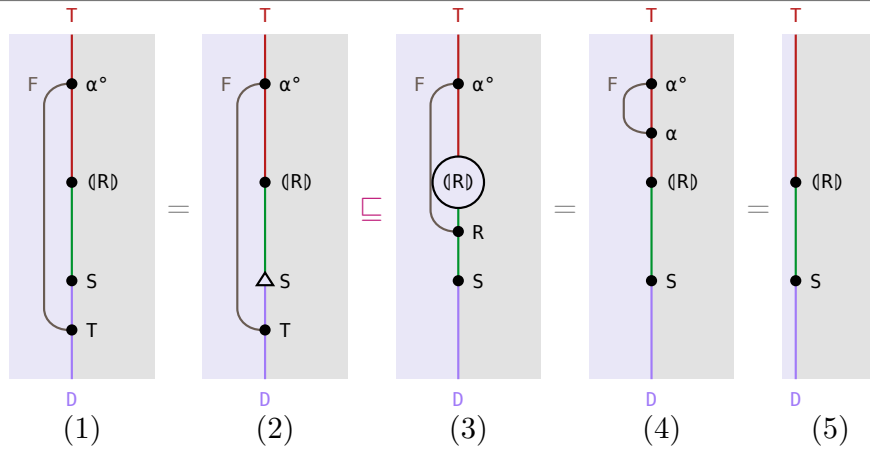
(6.2b)

the law	what it says
1 $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
2 $\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
3 $\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
4 $X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

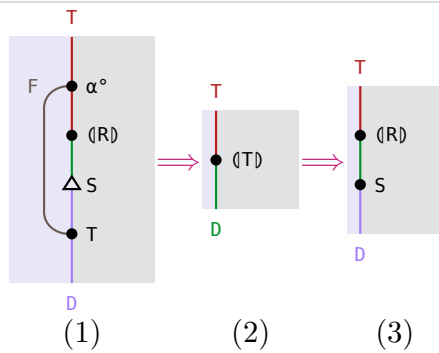
$$F(S) \ T \sqsubseteq R \ S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle \ S$$

(6.2c)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



1 $\alpha^\circ F(\langle R \rangle \ S) \ T$	
2 $= \alpha^\circ F(\langle R \rangle) \ F(S) \ T$	$F(S) \ T \sqsubseteq R \ S$
3 $\sqsubseteq \alpha^\circ F(\langle R \rangle) \ R \ S$	$\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$
4 $= \alpha^\circ \alpha \ \langle R \rangle \ S$	$\alpha^\circ \alpha = \mathbb{1}$
5 $= \langle R \rangle \ S$	

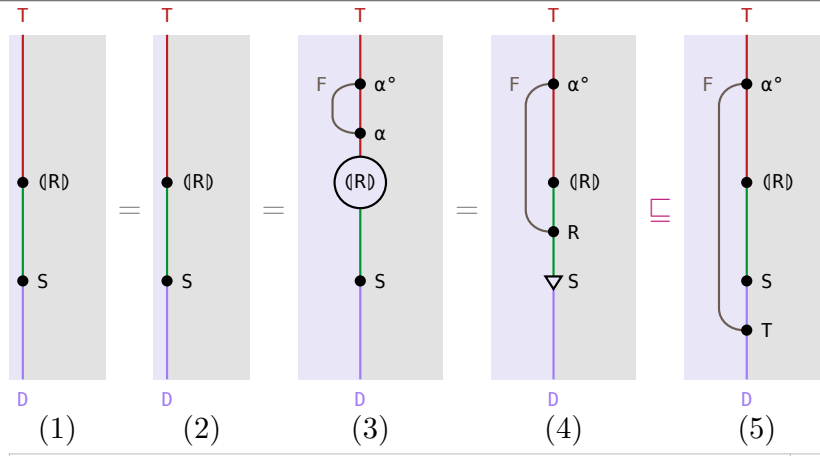


1 $F(S) \ T \sqsubseteq R \ S$	$F(S) \ T \sqsubseteq R \ S$ $\Rightarrow \alpha^\circ F(\langle R \rangle \ S) \ T \sqsubseteq \langle R \rangle \ S$
2 $\Rightarrow \alpha^\circ F(\langle R \rangle \ S) \ T \sqsubseteq \langle R \rangle \ S$	$\alpha^\circ F(X) \ R \sqsubseteq X$ $\Rightarrow \langle R \rangle \sqsubseteq X$
3 $\Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle \ S$	

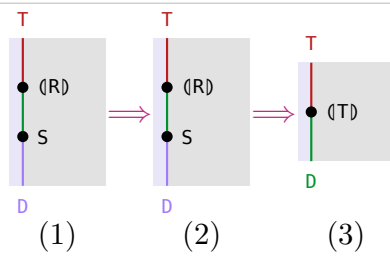
$$R \ S \sqsubseteq F(S) \ T \Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$$

(6.2d)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



1	$\langle R \rangle \ S$	
2	$= \langle R \rangle \ S$	$\alpha^\circ \ \alpha = 1$
3	$= \alpha^\circ \ \alpha \ \langle R \rangle \ S$	$\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$
4	$= \alpha^\circ \ F(\langle R \rangle) \ R \ S$	$R \ S \sqsubseteq F(S) \ T$
5	$\sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$	

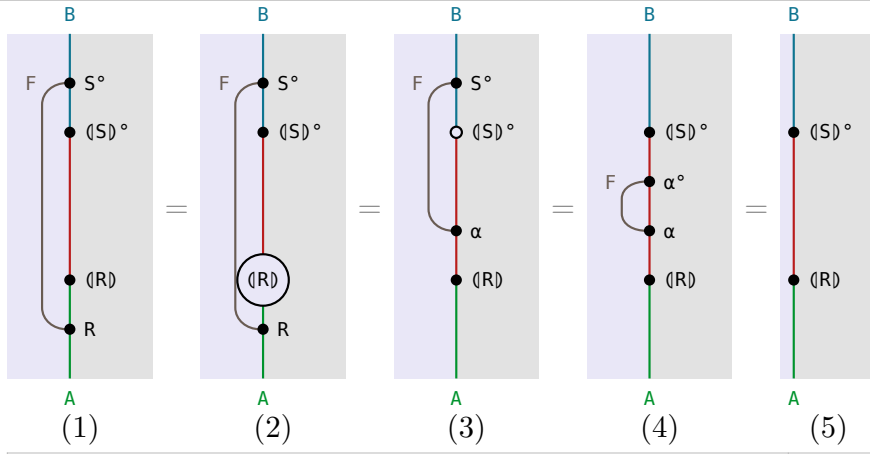


1	$R \ S \sqsubseteq F(S) \ T$	$R \ S \sqsubseteq F(S) \ T$ $\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$
2	$\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$	$X \sqsubseteq \alpha^\circ \ F(X) \ R$ $\Rightarrow X \sqsubseteq \langle R \rangle$
3	$\Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$	

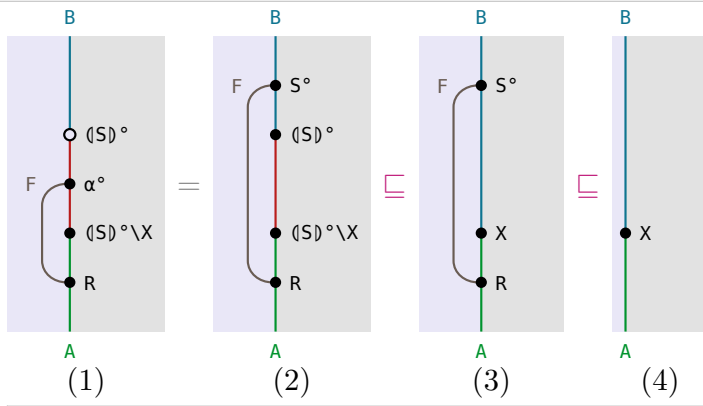
§6.3. Hylomorphisms

$$(S)^\circ (R) = (\mu X : S^\circ F(X) R) \quad (6.3a)$$

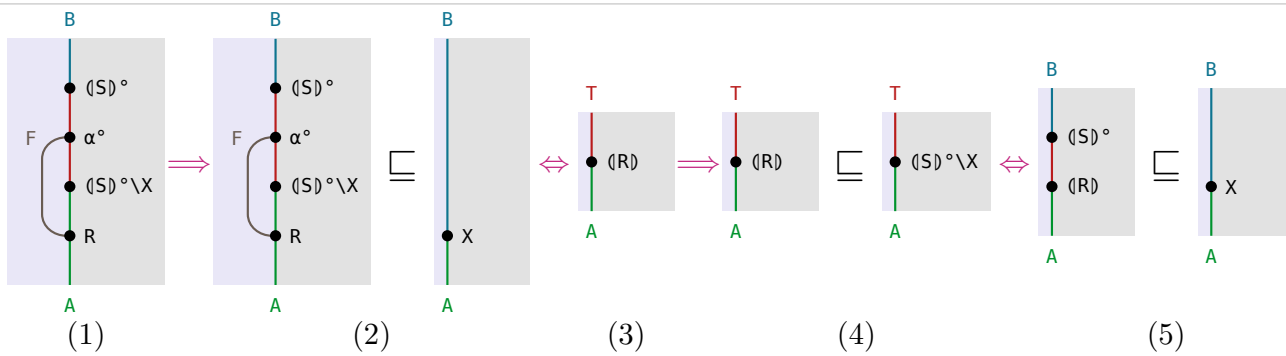
hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



1	$S^\circ F((S)^\circ (R)) R$	
2	$= S^\circ F((S)^\circ) F((R)) R$	$\alpha (R) = F((R)) R$
3	$= S^\circ F((S)^\circ) \alpha (R)$	$(R)^\circ \alpha^\circ = R^\circ F((R)^\circ)$
4	$= (S)^\circ \alpha^\circ \alpha (R)$	$\alpha^\circ \alpha = 1$
5	$= (S)^\circ (R)$	



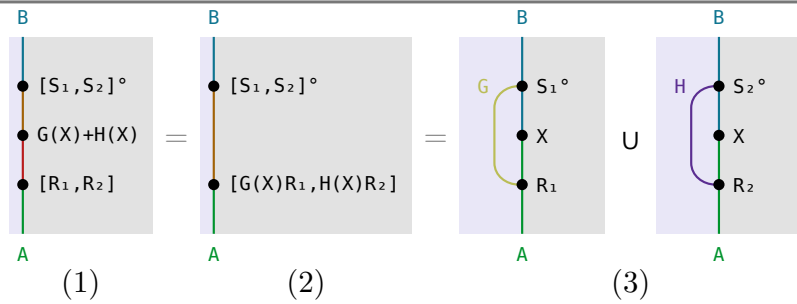
1	$(S)^\circ \alpha^\circ F((S)^\circ \setminus X) R$	$(R)^\circ \alpha^\circ = R^\circ F((R)^\circ)$
2	$= S^\circ F((S)^\circ ((S)^\circ \setminus X)) R$	$S(S \setminus R) \subseteq R$
3	$\sqsubseteq S^\circ F(X) R$	$S^\circ F(X) R \subseteq X$
4	$\sqsubseteq X$	



1	$S^\circ F(X) R \subseteq X$	$S^\circ F(X) R \subseteq X$
2	$\Rightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \subseteq X$	$\Rightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \subseteq X$
3	$\Leftrightarrow \alpha^\circ F((S)^\circ \setminus X) R \subseteq (S)^\circ \setminus X$	$X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$
4	$\Rightarrow (R) \subseteq (S)^\circ \setminus X$	$\alpha^\circ F(X) R \subseteq X$
5	$\Leftrightarrow (S)^\circ (R) \subseteq X$	$\Rightarrow (R) \subseteq X$

$$([S_1, S_2]^\circ)^\circ ([R_1, R_2])^\circ = (\mu X : (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results



1	$[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$	$(R + S) [P, Q] = [R P, S Q]$
2	$= [S_1, S_2]^\circ [G(X) R_1, H(X) R_2]$	$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$
3	$= (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)$	

§6.4. Fast exponentiation and modulus computation

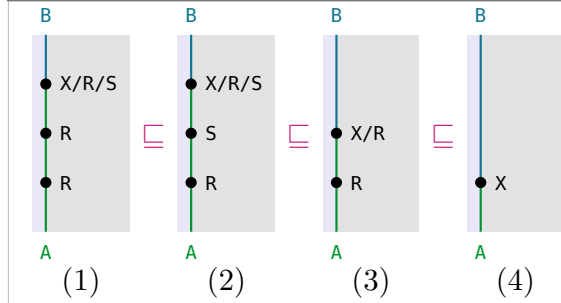
$\text{shift } f = (f \times 1) \ h \Rightarrow (\mu X : (\text{zero}^\circ \text{ zero } f) \cup (\text{shift}^\circ (X \times 1) h)) \sqsubseteq f$ once f turns shift into h, the recursion that halves the argument at each step computes f		(6.4a)
(1) (2) (3) (4)		
1	$(\mu X : (\text{zero}^\circ \text{ zero } f) \cup (\text{shift}^\circ (X \times 1) h))$	$\text{convert}^\circ ([g, h]) = (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$
2	$= \text{convert}^\circ ([\text{zero } f, h])$	$\text{zero } f = g \wedge \text{shift } f = (f \times 1) \ h$
3	$= \text{convert}^\circ \text{ convert } f$	$\Rightarrow \text{convert } f = ([g, h])$
4	$\sqsubseteq f$	$\text{Simple}(\text{convert})$

$(\mu X : (\text{zero}^\circ \text{ one}) \cup (\text{shift}^\circ (X \times 1) \text{ op}(a))) \sqsubseteq \text{exp}(a)$ the recursion that halves the exponent at each step computes a to the power b		(6.4b)
$\text{zero exp}(a) = \text{one}$ a to the power 0 is 1		
$\text{shift exp}(a) = (\text{exp}(a) \times 1) \text{ op}(a)$ a to the power $2n+d$ is $\text{op}(a)$ of a to the power n and d		

$(\mu X : (\text{zero}^\circ \text{ zero}) \cup (\text{shift}^\circ (X \times 1) \text{ op}(b))) \sqsubseteq \text{mod}(b)$ the recursion that halves a at each step computes the remainder of a by b		(6.4c)
$\text{zero mod}(b) = \text{zero}$ $0 \text{ mod } b$ is 0		
$\text{shift mod}(b) = (\text{mod}(b) \times 1) \text{ op}(b)$ $(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and d		

§6.5. Unique fixed points

$$\begin{aligned}
 & \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{X} & (6.5a) \\
 & ((\mathbf{X}/\mathbf{R})/\mathbf{S}) \text{ followed by } \mathbf{R}\mathbf{R} \text{ stays below } \mathbf{X} \text{ when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \\
 & \mathbf{inductive}(\mathbf{S}) \wedge \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow \mathbf{inductive}(\mathbf{R}) \\
 & \mathbf{R} \text{ is inductive when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \text{ for an inductive } \mathbf{S}
 \end{aligned}$$



1	$(\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R}$	$\mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R}$
2	$\sqsubseteq (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{S} \mathbf{R}$	$(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$
3	$\sqsubseteq (\mathbf{X} / \mathbf{R}) \mathbf{R}$	$(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$
4	$\sqsubseteq \mathbf{X}$	

$$\mathbf{member} = \emptyset \quad (6.5b)$$

a constant relator records no elements, so its membership is empty

$$\mathbf{member} = [\mathbf{member}, \mathbf{member}]$$

a member of $\mathbf{F} + \mathbf{G}$ is a member of whichever summand the value is in

$$\mathbf{member} = (\pi_1 \mathbf{member}) \cup (\pi_2 \mathbf{member})$$

a member of $\mathbf{F} \times \mathbf{G}$ is a member of either component

$$\mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{X} = \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \wedge \mathbf{Y} = \mathbf{S} \mathbf{F}(\mathbf{Y}) \mathbf{R} \Rightarrow \mathbf{X} = \mathbf{Y} \quad (6.5c)$$

when $\mathbf{S} \mathbf{member}$ is inductive, two solutions $\mathbf{X}, \mathbf{Y}$ of $\mathbf{X} = \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R}$ are equal

$$\mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{Entire}(\mathbf{S}) \wedge \mathbf{Entire}(\mathbf{R}) \wedge \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \sqsubseteq \mathbf{X} \Rightarrow \mathbf{Entire}(\mathbf{X}) \quad (6.5d)$$

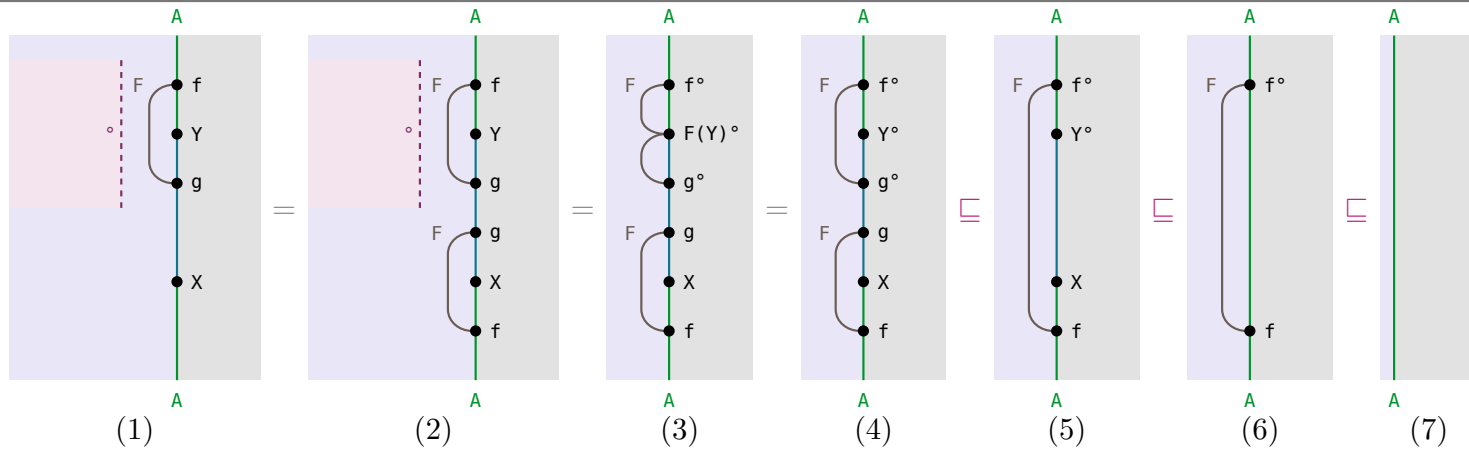
when $\mathbf{S} \mathbf{member}$ is inductive and $\mathbf{S}, \mathbf{R}$ are entire, a pre-fixed point of $\mathbf{X} \mapsto \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R}$ is entire

$$\mathbf{inductive}(\mathbf{g} \mathbf{member}) \wedge \mathbf{Map}(\mathbf{g}) \wedge \mathbf{Map}(\mathbf{f}) \wedge \mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \Rightarrow \mathbf{Map}(\mathbf{X}) \quad (6.5e)$$

when $\mathbf{g} \mathbf{member}$ is inductive and $\mathbf{f}, \mathbf{g}$ are maps, the solution of $\mathbf{X} = \mathbf{g}\mathbf{F}(\mathbf{X})\mathbf{f}$ is a map

$$\mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \wedge \mathbf{Simple}(\mathbf{g}) \wedge \mathbf{Simple}(\mathbf{f}) \wedge \mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1} \Rightarrow (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X} \sqsubseteq \mathbf{1}$$

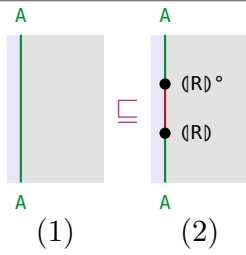
$(\mathbf{g}\mathbf{F}(\mathbf{Y})\mathbf{f})^\circ$ followed by $\mathbf{X}$ stays below $\mathbf{1}$ when $\mathbf{g}, \mathbf{f}$ are simple and $\mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}$



1	$(\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X}$	$\mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$
2	$= (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$	$(\mathbf{R} \mathbf{S})^\circ = \mathbf{S}^\circ \mathbf{R}^\circ$
3	$= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y})^\circ \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$	$\mathbf{PreservesRecip}(\mathbf{F})$
4	$= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ) \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$	$\mathbf{Simple}(\mathbf{g})$
5	$\sqsubseteq \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ \mathbf{X}) \mathbf{f}$	$\mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}$
6	$\sqsubseteq \mathbf{f}^\circ \mathbf{f}$	$\mathbf{Simple}(\mathbf{f})$
7	$\sqsubseteq \mathbf{1}$	

$$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \Rightarrow 1 \sqsubseteq (R)^\circ (R) \quad (6.5f)$$

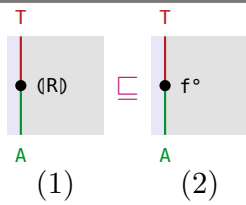
when $R^\circ \text{member}$ is inductive and R is surjective, the fold (R) is surjective



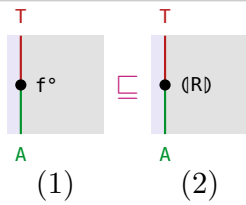
1	1	$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S$ $F(X) R \sqsubseteq X$ $\Rightarrow \text{Entire}(X) \text{ at } S=R^\circ$
2	$\sqsubseteq (R)^\circ (R)$	

$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = (R) \quad (6.5g)$$

a map f with $Rf \sqsubseteq F(f)\alpha$, for a surjective R , has the fold (R) as its converse



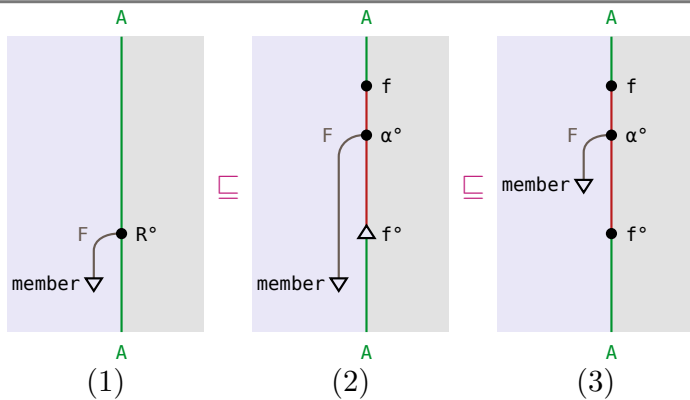
1	(R)	shunting f , fusion, $Rf \sqsubseteq F(f)\alpha$
2	$\sqsubseteq f^\circ$	



1	f°	$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R$ $\Rightarrow 1 \sqsubseteq (R)^\circ (R) \text{ and } (R)f \sqsubseteq 1$
2	$\sqsubseteq (R)$	

$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \Rightarrow R^\circ \text{ member} \sqsubseteq f \alpha^\circ \text{ member } f^\circ \quad (6.5h)$$

$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



1	$R^\circ \text{ member}$	$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha$ $\Rightarrow R^\circ \sqsubseteq f \alpha^\circ F(f^\circ)$
2	$\sqsubseteq f \alpha^\circ F(f^\circ) \text{ member}$	$F(R) \text{ member} \sqsubseteq \text{member } R$
3	$\sqsubseteq f \alpha^\circ \text{ member } f^\circ$	

§6.6. Sorting by selection

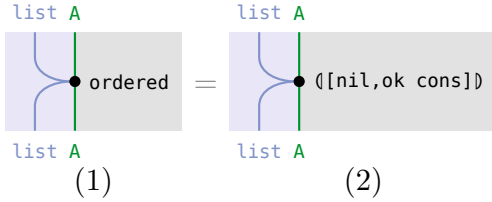
(6.6a)

	name	type	definition	meaning
1	orderedP(R)	P[A]	$\text{orderedP}[] = \text{true}, \text{orderedP}([a]\#xs) = (\forall x. x \in xs \Rightarrow R(a,x)) \wedge \text{orderedP}(xs)$	every element is R-below each element after it
2	ordered	$[A] \rightarrow [A]$	$xs \text{ ordered } ys \Leftrightarrow xs = ys \wedge \text{orderedP}(xs)$	the coreflexive passing exactly the lists orderedP holds of
3	ok	$A \times [A] \rightarrow A \times [A]$	$(a,xs) \text{ ok } q \Leftrightarrow (a,xs) = q \wedge (\forall x. x \in xs \Rightarrow R(a,x))$	the coreflexive passing (a, x) when a is R-below every element of x

ordered = ([nil,ok cons])

(6.6b)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

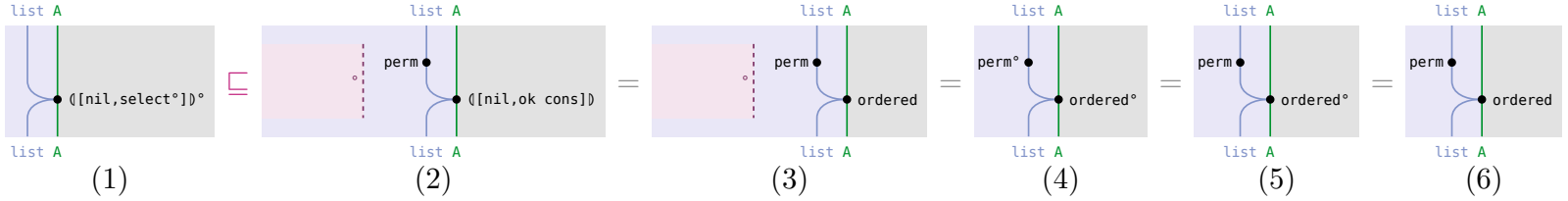


1	ordered	fold uniqueness
2	= ([nil,ok cons])	

select° ⊆ ok cons perm ⇒ ([nil,select°])° ⊆ perm ordered

(6.6c)

every output of unfolding the input by **select** is a sorted permutation of it

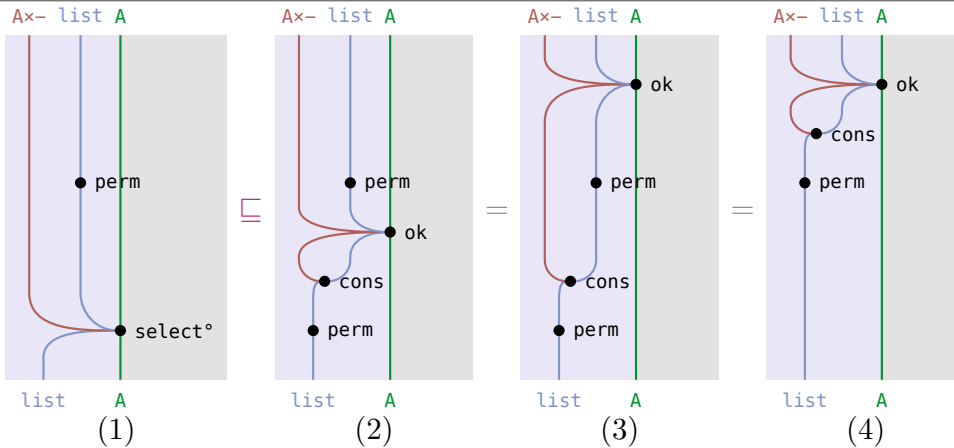


1	$([nil,select^\circ])^\circ$	$select^\circ \sqsubseteq ok \ cons \ perm$
2	$\sqsubseteq ([nil,ok \ cons])^\circ \ perm)^\circ$	$\Rightarrow ([nil,select^\circ])^\circ \sqsubseteq ([nil,ok \ cons])^\circ \ perm)^\circ$
3	$= (ordered \ perm)^\circ$	(6.6b) $ordered = ([nil,ok \ cons])$
4	$= perm^\circ \ ordered^\circ$	$(R \ S)^\circ = S^\circ \ R^\circ$
5	$= perm \ ordered^\circ$	$perm^\circ = perm$
6	$= perm \ ordered$	$ordered^\circ = ordered$

select° ⊆ ok cons perm ⇒ (1 × perm) select° ⊆ ok cons perm

(6.6d)

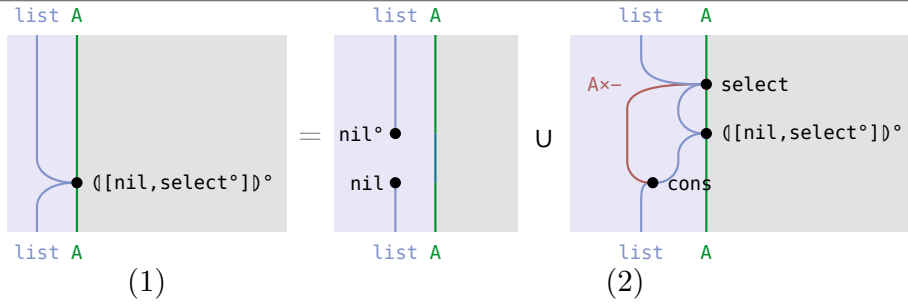
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



1	$(1 \times perm) \ select^\circ$	$select^\circ \sqsubseteq ok \ cons \ perm$
2	$\sqsubseteq (1 \times perm) \ ok \ cons \ perm$	$ok \ (1 \times perm) = (1 \times perm) \ ok$
3	$= ok \ (1 \times perm) \ cons \ perm$	$cons \ perm = (1 \times perm) \ cons \ perm$
4	$= ok \ cons \ perm$	

$$([nil, select^\circ])^\circ = (nil^\circ nil) \cup (select (1 \times ([nil, select^\circ])^\circ) cons) \quad (6.6e)$$

selection sort returns [] on [], and otherwise selects (a,y), sorts y and conses a back on

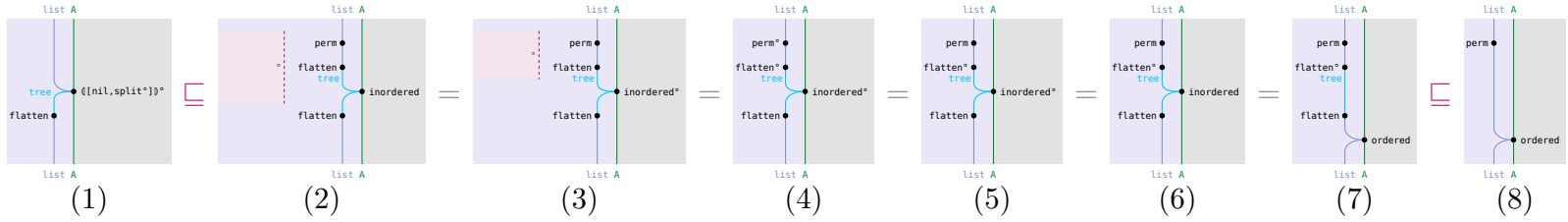


1	$[nil, select^\circ]^\circ$	unfolding the converse of a fold
2	$= (nil^\circ nil) \cup (select (1 \times ([nil, select^\circ])^\circ) cons)$	

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge split^\circ \sqsubseteq check' \text{ join perm} \Rightarrow ([nil, split^\circ])^\circ \quad (6.6f)$$

flatten $\sqsubseteq$ perm ordered

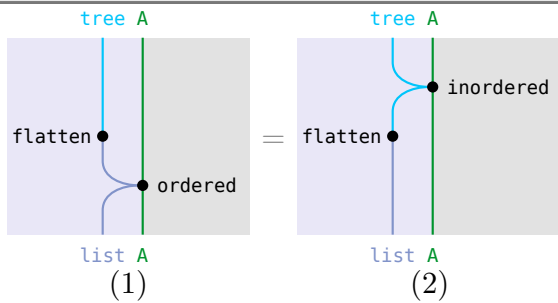
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



1	$[nil, split^\circ]^\circ$ flatten	$split^\circ \sqsubseteq check' \text{ join perm}$
2	$\sqsubseteq (inordered \text{ flatten } perm)^\circ$ flatten	$\Rightarrow ([nil, split^\circ])^\circ \text{ flatten} \sqsubseteq (inordered \text{ flatten } perm)^\circ$
3	$= (flatten \text{ perm})^\circ$ inordered $^\circ$ flatten	$(R S)^\circ = S^\circ R^\circ$
4	$= perm^\circ flatten^\circ$ inordered $^\circ$ flatten	$(R S)^\circ = S^\circ R^\circ$
5	$= perm \text{ flatten}^\circ$ inordered $^\circ$ flatten	$perm^\circ = perm$
6	$= perm \text{ flatten}^\circ$ inordered flatten	$inordered^\circ = inordered$
7	$= perm \text{ flatten}^\circ$ flatten ordered	$(6.6g) (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c))$
8	$\sqsubseteq perm \text{ ordered}$	$\Rightarrow \text{flatten ordered} = \text{inordered flatten}$
		$flatten^\circ \text{ flatten} \sqsubseteq 1$

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6g)$$

flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening

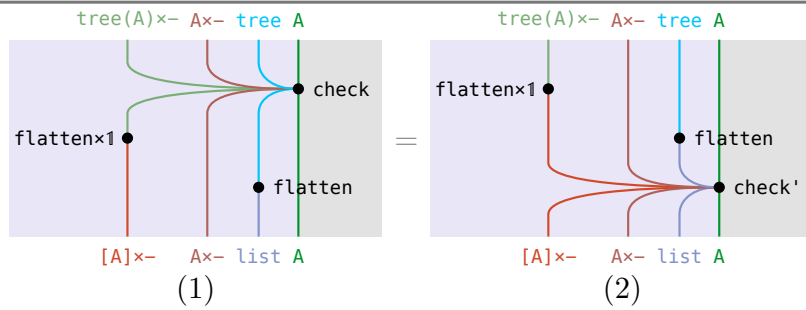


1	$flatten \text{ ordered}$	R transitive: a flattened tree is ordered iff every node passes check
2	$= inordered \text{ flatten}$	

check (flatten × (1 × flatten)) = (flatten × (1 × flatten)) check'

(6.6h)

testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening

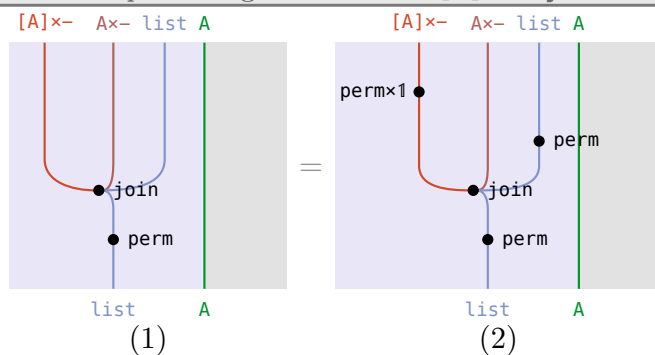


1	check (flatten × (1 × flatten))	b intree x ⇔ b inlist flatten(x)
2	= (flatten × (1 × flatten)) check'	

join perm = (perm × (1 × perm)) join perm

(6.6i)

permuting the list **x ++ [a] ++ y** is the same as first permuting **x** and **y** and then permuting the joined list

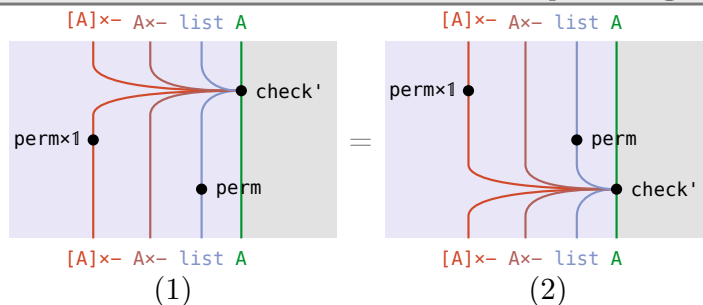


1	join perm	perm is transitive and closed under ++
2	= (perm × (1 × perm)) join perm	

check' (perm × (1 × perm)) = (perm × (1 × perm)) check'

(6.6j)

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

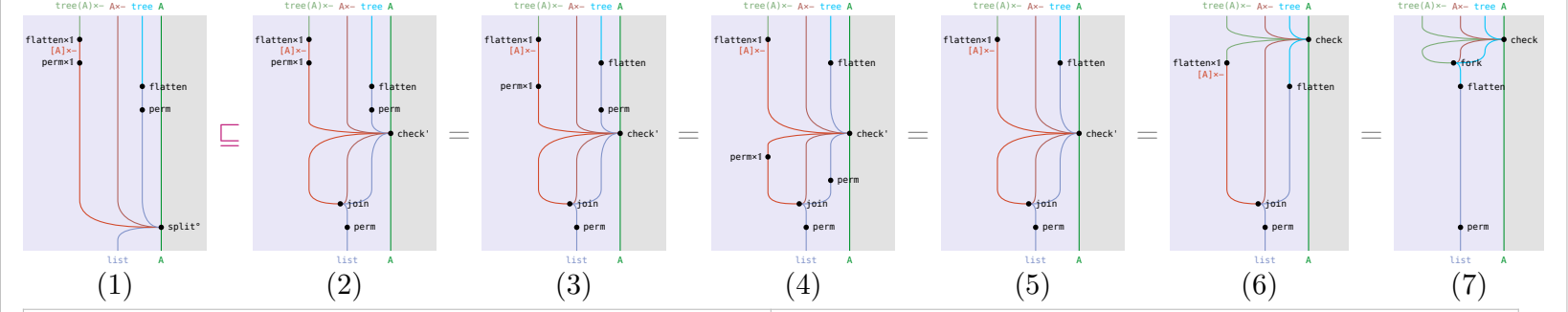


1	check' (perm × (1 × perm))	b inlist x ⇔ b inlist perm(x)
2	= (perm × (1 × perm)) check'	

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ \sqsubseteq \text{check fork} \quad (6.6k)$$

flatten perm

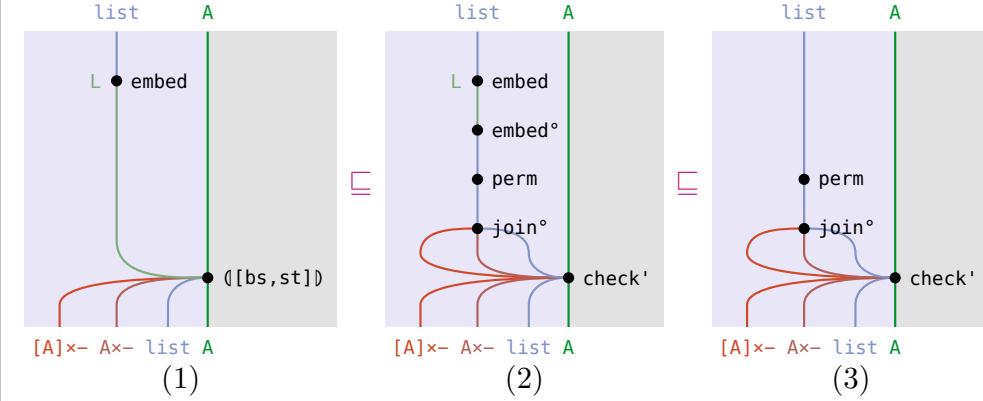
undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is **permuted**



1	$((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ$	$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$
2	$\sqsubseteq ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ check}' \text{ join perm}$	
3	$= (\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{ check}' \text{ join perm}$	(6.6j) $\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}'$
4	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$	(6.6i) $\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$
5	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \text{ join perm}$	(6.6h) $\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$
6	$= \text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{ join perm}$	$\text{fork flatten} = (\text{flatten} \times (1 \times \text{flatten})) \text{ join}$
7	$= \text{check fork flatten perm}$	

$$\text{bs} \sqsubseteq \text{wrap perm join}^\circ \text{ check}' \wedge (1 \times (\text{perm join}^\circ \text{ check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{ check}' \Rightarrow \text{embed} \quad (6.6l)$$

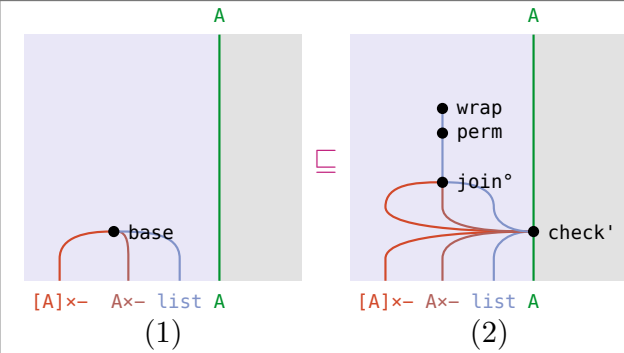
turning a list into a non-empty list and folding it with any $[\text{bs}, \text{st}]$ meeting the two conditions below gives only **checked triples** (x, a, y) whose $\text{join } x \# [a] \# y$ is a permutation of the list



1	$\text{embed } ([\text{bs}, \text{st}])$	$\text{bs} \sqsubseteq \text{wrap perm join}^\circ \text{ check}' \wedge (1 \times (\text{perm join}^\circ \text{ check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{ check}'$ $\Rightarrow \text{embed } ([\text{bs}, \text{st}]) \sqsubseteq \text{embed embed}^\circ \text{ perm join}^\circ \text{ check}'$
2	$\sqsubseteq \text{embed embed}^\circ \text{ perm join}^\circ \text{ check}'$	$\text{embed embed}^\circ \text{ perm join}^\circ \text{ check}' \sqsubseteq \text{perm join}^\circ \text{ check}'$
3	$\sqsubseteq \text{perm join}^\circ \text{ check}'$	

$$\text{base} \sqsubseteq \text{wrap perm join}^\circ \text{ check}' \quad (6.6m)$$

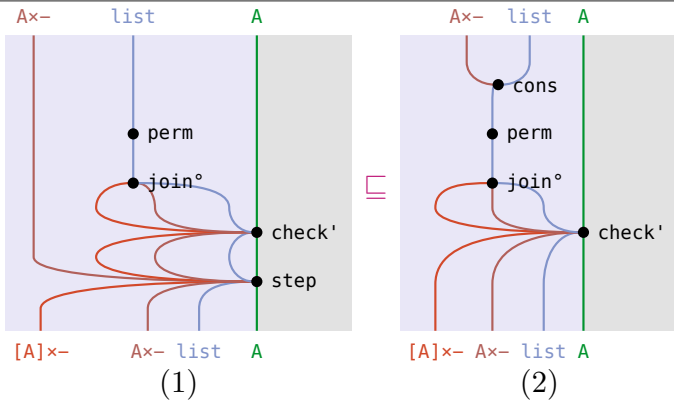
base(a) is a **checked triple** whose **join** is a permutation of the one-element list $[a]$



1	base	$[\] \# [a] \# [\] = [a]$, and check' holds on empty lists
2	$\sqsubseteq \text{wrap perm join}^\circ \text{ check}'$	

$$\text{connected}(R) \Rightarrow (1 \times (\text{perm join}^\circ \text{check}')) \text{ step} \sqsubseteq \text{cons perm join}^\circ \text{check}' \quad (6.6n)$$

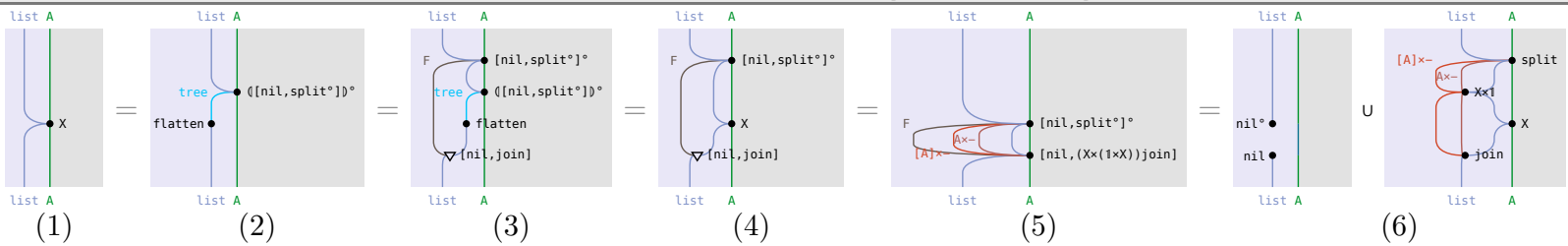
if (x, b, y) is a **checked** triple for a permutation of l , then $\text{step}(a, (x, b, y))$ is a **checked** triple for a permutation of $[a] \# l$



1	$(1 \times (\text{perm join}^\circ \text{check}')) \text{ step}$	aRb puts a in front of x ; otherwise bRa and a goes in front of y
2	$\sqsubseteq \text{cons perm join}^\circ \text{check}'$	

$$X = ([\text{nil}, \text{split}^\circ])^\circ \text{flatten} \Rightarrow X = (\text{nil}^\circ \text{nil}) \cup (\text{split} (X \times (1 \times X)) \text{join}) \quad (6.6o)$$

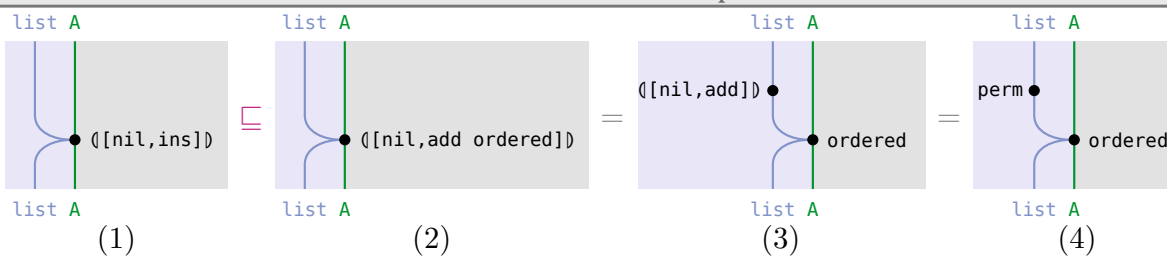
quicksort returns $[]$ on $[]$, and otherwise splits into (x, a, y) , sorts x and y and joins them around a



1	X	$X = ([\text{nil}, \text{split}^\circ])^\circ \text{flatten}$
2	$= ([\text{nil}, \text{split}^\circ])^\circ \text{flatten}$	$S^\circ F((S)^\circ (R)) R = (S)^\circ (R)$
3	$= [\text{nil}, \text{split}^\circ]^\circ F([\text{nil}, \text{split}^\circ]^\circ \text{flatten}) [\text{nil}, \text{join}]$	$X = ([\text{nil}, \text{split}^\circ])^\circ \text{flatten}$
4	$= [\text{nil}, \text{split}^\circ]^\circ F(X) [\text{nil}, \text{join}]$	$F(S) [T, U] = [T, (S \times (1 \times S)) U]$
5	$= [\text{nil}, \text{split}^\circ]^\circ [\text{nil}, (X \times (1 \times X)) \text{join}]$	$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$
6	$= (\text{nil}^\circ \text{nil}) \cup (\text{split} (X \times (1 \times X)) \text{join})$	

$$(1 \times \text{ordered}) \text{ins} \sqsubseteq \text{add ordered} \Rightarrow ([\text{nil}, \text{ins}]) \sqsubseteq \text{perm ordered} \quad (6.6p)$$

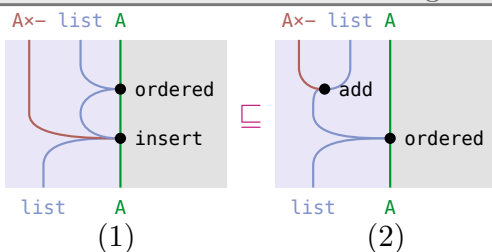
for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with $[\text{nil}, \text{ins}]$ gives only sorted permutations



1	$[\text{nil}, \text{ins}]$	$(1 \times \text{ordered}) \text{ins} \sqsubseteq \text{add ordered}$
2	$\sqsubseteq [\text{nil}, \text{add ordered}]$	$\Rightarrow ([\text{nil}, \text{ins}]) \sqsubseteq ([\text{nil}, \text{add ordered}])$
3	$= [\text{nil}, \text{add}] \text{ordered}$	$[\text{nil}, \text{add}] \text{ordered} = [\text{nil}, \text{add ordered}]$
4	$= \text{perm ordered}$	$\text{perm} = [\text{nil}, \text{add}]$

$$\text{connected}(R) \wedge (\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \Rightarrow (1 \times \text{ordered}) \text{insert} \sqsubseteq \text{add ordered} \quad (6.6q)$$

inserting a into an ordered list puts a somewhere in it and keeps it ordered



1	$(1 \times \text{ordered}) \text{insert}$	insert splices a in, and R transitive and connected
2	$\sqsubseteq \text{add ordered}$	

§6.7. Closure

$$R^* = (\mu X : 1 \cup (X R)) \quad (6.7a)$$

closing $\mathbf{R}$ by composing it on the left and closing it by composing on the right give the same relation $\mathbf{R}^*$

$$1 \in X \wedge X \subseteq X \Rightarrow R \subseteq X \Leftrightarrow R^* \subseteq X \quad (6.7b)$$

X contains R exactly when it contains R*

$$1 \in \mathbb{R}^* \quad (6.7c)$$

R^* is reflexive

1	1	$R \subseteq R \cup S$
2	$\subseteq 1 \cup (R \cup R^*)$	$1 \cup (R \cup R^*) = R^*$
3	$= R^*$	

$$\mathbf{R} \in \mathbf{R}^* \quad (6.7d)$$

R^* contains R

1	R	
2	$= R$	$(6.7c) \quad 1 \subseteq R^*$
3	$\subseteq R \ R^*$	$R \ R^* \subseteq R^*$
4	$\subseteq R^*$	

$$(1 \cup (R(R^* / R^*))) R^* \subseteq R^* \quad (6.7e)$$

R^*/R^* is a prefixed point of $X \mapsto 1 \cup R X$, stated through division

$$R^* \cap R^* \subseteq R^*$$

R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$

1	$(1 \cup (R \setminus R^*)) \cup R^*$	$(S \cup T) \cap R = (S \cap R) \cup (T \cap R)$
2	$= (R^*) \cup (R \setminus R^*)$	
3	$= R^* \cup (R \setminus R^*)$	$(R \setminus R) \cap R \subseteq R$
4	$\subseteq R^* \cup (R \cap R^*)$	$R \cap R^* \subseteq R^*$
5	$\subseteq R^*$	

$$1 \in X \wedge X \subseteq X \wedge R \subseteq X \Rightarrow 1 \cup (R \setminus X) \subseteq X \quad (6.7f)$$

a reflexive transitive $\mathbf{X}$ containing $\mathbf{R}$ is a prefixed point of $\mathbf{Y} \mapsto 1 \cup \mathbf{R}\mathbf{Y}$

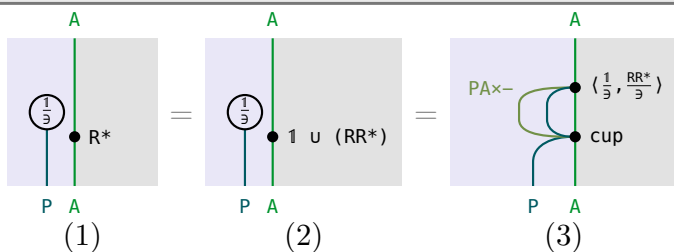
$$1 \in X \wedge X \subseteq X \wedge R \subseteq X \implies R^* \subseteq X$$

a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup RX$, so it contains R^*

1	$1 \cup (R \times X)$	$1 \subseteq X$
2	$\subseteq X \cup (R \times X)$	$R \subseteq X$
3	$\subseteq X \cup (X \times X)$	
4	$\subseteq X$	

$$\frac{R^*}{\vartheta} = \left\langle \frac{1}{\vartheta}, \frac{R}{\vartheta} \frac{R^*}{\vartheta} \right\rangle \text{ cup} \quad (6.7g)$$

the set of $\mathbf{R}^*$ -successors of $\mathbf{a}$ is $\mathbf{a}$ itself joined with the $\mathbf{R}^*$ -successors of its $\mathbf{R}$ -successors



1	$\frac{R^*}{\exists}$	$1 \cup (R \cup R^*) = R^*$
2	$= \frac{1 \cup (R \cup R^*)}{\exists}$	$\frac{R \cup S}{\exists} = \langle \frac{R}{\exists}, \frac{S}{\exists} \rangle \text{ cup}$
3	$= \langle \frac{1}{\exists}, \frac{R \cup R^*}{\exists} \rangle \text{ cup}$	

$R - \emptyset = R$	(6.7h)
taking nothing away leaves R	
$R \cup (S - R) = R \cup S$	
beside R , only the part of S outside R adds anything	
$R - (S \cup T) = R - S - T$	
taking away S and then T takes away $S \cup T$	
$(R \cup S) - T = (R - T) \cup (S - T)$	
taking T away from a union takes it away from each part	

$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$	(6.7i)
the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ	

$S R^* = (\mu X : S \cup (X R))$	(6.7j)
S followed by any number of R steps is the least X containing S and closed under a further R step	
$R^* S = (\mu X : S \cup (R X))$	
any number of R steps followed by S is the least X containing S and closed under an R step in front	

$\theta(\emptyset, S) = S R^*$	(6.7k)
started with nothing found and S to explore, θ computes $S R^*$	
1 $\theta(\emptyset, S)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
2 $\triangleq \emptyset \cup (\mu X : S \cup ((X R) - \emptyset))$	(6.7h) $R - \emptyset = R$
3 $= \emptyset \cup (\mu X : S \cup (X R))$	$\emptyset \cup R = R$
4 $= (\mu X : S \cup (X R))$	(6.7j) $S R^* = (\mu X : S \cup (X R))$
5 $= S R^*$	

$\theta(P, \emptyset) = P$	(6.7l)
with nothing left to explore, θ returns what it has found	
1 $\theta(P, \emptyset)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
2 $\triangleq P \cup (\mu X : \emptyset \cup ((X R) - P))$	$(\mu X : \emptyset \cup ((X R) - P)) = \emptyset$
3 $= P \cup \emptyset$	$R \cup \emptyset = R$
4 $= P$	